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Sustainable Development Indicators: From Statistics to Policy

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ABSTRACT

Sustainable development indicators (SDIs) may have good potential to bring environmental concerns to the policy agenda. However, different understandings of sustainability, definitions of SDIs and measurement procedures may give completely different assessments of whether society moves towards a sustainable development path or not. Compilation of statistical indicators for environmental change and sustainability comprises not only a selection of facts in some technical sense, as the choices involved are conditioned by societal interests and implicit values embedded in the data-generating processes. This implies that statistical offices cannot ignore the role that values play in the generation of accurate data sets. To give an assessment of sustainability, we need not only to address historical trends but also to evaluate policy choices made today and how they may influence future development. SDI sets should be evaluated according to how they contribute to deliberation on sustainability in learning processes involving participants beyond the science–policy interface. Copyright © 2012 John Wiley & Sons, Ltd and ERP Environment.

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Introduction

ALTHOUGH THE TERM SUSTAINABILITY ADDRESSES THE CONTINUITY AND STABILITY OF HUMANITY'S FUTURE ON THIS planet, it can be argued that the possibility of reaching a sustainable development rests in a commitment to the present, reflected in political action, as well as in accountability to civil society (Benessia *et al.*, 2012). This raises the issue of quality control of knowledge claims associated with quantitative analysis of how to improve the sustainability of our societies (Jasanoff, 2005). Normative (value-driven) choices can substantially bias assessments of whether society moves towards a sustainable development path or not, in terms of: (1) choice of criteria associated with different understandings of sustainability; (2) choice of narratives regarding sustainability, leading to different definitions of sustainable development indicators (SDIs) based on different attributes of performance; and (3) choice of how to quantify the chosen attributes in terms of quantitative variables

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(indicators) and statistical methods, as for instance sampling procedures. Environmental concerns call for multi-dimensional SDI sets, in order to express strong (ecological) sustainability and incommensurable values (Ramsteiner *et al.*, 2007; Stiglitz *et al.*, 2009; Nilsen, 2010).

Here we discuss how SDI sets in general are used in assessments of whether society moves towards a sustainable development path or not, and to what extent there is feedback taking place between statistics and policy. A comprehensive survey of indicators and assessment procedures is beyond the scope of this paper. Our aim is to critically discuss some widely applied procedures for sustainability assessment and consider the key role of national statistical offices and other institutions of society for improving the knowledge basis for policy for sustainable development, an issue that is also at the core of the Stiglitz report (Stiglitz *et al.*, 2009).

The research question we focus on is how the assessment of sustainability, taking place along a path from statistics to policy, can be improved by taking into account the importance of implicit value assumptions and fundamental uncertainties (Sarewitz, 2004). We introduce an organizational learning model as a tool for improving the assessment process. To express societal differences in values, perspectives and interests, we apply the concept of narrative, i.e. the "story" behind the data. We discuss the importance of fundamental uncertainty and take a critical look at the call for composite sustainability indicators. Finally, we conclude by considering the application of SDIs for policy. Our main findings are that: (1) clarification of the learning processes and their normative context is crucial to improve the assessment procedures, (2) the stepwise procedure from indicators to assessment and policy needs to be extended to make the narratives of sustainability visible in the policy process (Figure 5) and (3) sustainability indicators and assessments can be important tools for policy. The main message is that it is not only the overall quality of the indicators or the general model applied that matters, but rather how these data and assessments are deliberated in a political process reaching agreements for political action.

The Standard Sustainability Assessment Process

The standard approach to sustainability assessment may be described as a sequential process (Figure 1). The selection and definition of the SDI set (A) and the formulation of SDI targets and policy actions (D) are normally the responsibility of policy-makers and follow as result of a political process. In some countries, the national statistical office is responsible for both the annual production of data for the SDI set (B) and the annual assessment of whether the SDI set points towards a sustainable development path or not (C). In other countries, SDI production is done by the national statistical office, while the assessment of sustainability is performed by a ministry or is part of a political process.

Standard indicator-based assessment of sustainability (C) also consists of several steps. The first step would typically be an assessment of a particular issue of sustainability based on a single indicator; for example, one for climate change may be expressed by data for greenhouse gas emissions. The information provided by a particular indicator may be supplemented by other statistical information or indicators. In the context of climate change, the choice of indicators may reflect different approaches to climate policy or different perspectives on how climate policy is perceived by the public and what policy options are perceived as available.

The sequential process in Figure 1 is reflected in the many attempts to create general frameworks for SDIs, including those proposed by the United Nations, Organisation for Economic Co-operation and Development (OECD), European Union (EU), World Bank or regional organizations (United Nations, 2007). In 2009, the International Institute for Sustainable Development reported some 800 different initiatives for indicator frameworks (International Institute for Sustainable Development, 2009).

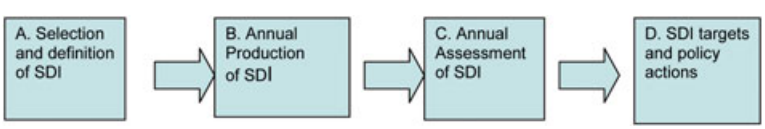


Figure 1. Sequential process for production and assessment of SDIs

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The standard situation of sustainability assessment is that there is an SDI set (sometimes a large set) typically categorized as economic, environmental and social indicators. For each indicator one can assess a direction of development which may be considered in line with a sustainable development path. The difficult question is, however, whether the total picture points toward a sustainable development path or not. Moreover, reports on sustainability assessments hardly provide any reflections on how the indicators depend upon each other or how they may be linked to each other and how trade-offs are to be evaluated. An exception is the capital approach, suggested on the grounds that seen together, the indicators may tell more about the potential for the future wealth of society than each indicator does on an individual basis (United Nations, 2009, 2011). However, the concept of national wealth rests on the assumption of substitution between incommensurable values, and the prices called for may not reflect the environmental and societal values or may in many cases be impossible to estimate. Due to this criticism, it is becoming widely accepted that the capital approach has to be expanded to include multidimensional environmental indicators as a basis for policy. The Stiglitz report has taken steps to meet some of this critique and argues that sustainability assessment requires "a dashboard of indicators":

> "The assessment of sustainability is complementary to the question of current well-being or economic performance, and must be examined separately. This may sound trivial and yet it deserves emphasis, because some existing approaches fail to adopt this principle, leading to potentially confusing messages. For instance, confusion may arise when one tries to combine current well-being and sustainability into a single indicator. To take an analogy, when driving a car, a meter that added up into one single number the current speed of the vehicle and the remaining level of gasoline would not be of any help to the driver. Both pieces of information are critical and need to be displayed in distinct, clearly visible areas of the dashboard.. [. . .] sustainability requires the simultaneous preservation or increase in several 'stocks': quantities and qualities of natural resources and of human, social and physical capital" (Stiglitz *et al.*, 2009, p. 17).

A practical approach to express interdependence and trade-offs between indicators, and provide reflections and interpretations of the indicators and narratives involved, may be to use radar charts to display multivariate indicators, as for example adopted by ten Brink *et al.* (1991) in the AMOEBA approach for analysis of ecological indicators for the North Sea. The idea of radar charts is to display multi-dimensional information. By careful selection of the sequence of the indicators presented, the radar chart may also be used to indicate the chronology of events, or the ordering of a phenomenon along a given dimension, helping to clarify and understand possible causal relationships between the indicators (Gomiero and Giampietro, 2005). The importance of different perspectives, reflected in the choice of, or weighing between, different indicators, may be illustrated on a relative scale as shown in the radar chart in Figure 2, where positioning in the outer field reflects high importance, the middle field reflects medium importance and the inner field reflects low importance. Figure 2 illustrates in a radar chart how the choice of indicators for climate policy can be seen as expressions of different science–policy perspectives that emphasize different framings and approaches to climate policy.

From a purely economic perspective, the most important aspects related to climate policy would typically be emission trading systems and taxes, climate agreements, and technology. Thus, the indicators selected would accordingly represent this perspective as indicated in Figure 2. From the perspective of integrated assessment, or the viewpoint of an ecological economist, the choice of climate policy indicators would typically include how climate policy is perceived by the public and what policy options are perceived as available and direct emphasis on environmental indicators such as ice cover in the Arctic and ocean acidification. The point here is not to construct dividing lines between economic, ecological or sociological indicators, but to emphasize the need for a more comprehensive set of indicators for assessment of climate policy, reflecting a plurality of normative value assumptions expressed in narratives. The recent financial crisis has led to renewed attention of the economics profession in furthering the importance of a plurality of perspectives for shaping policy discussions (Krugman, 2011). Different approaches to sustainability indicators may be seen as different fragments of a bigger picture that needs the connecting parts filled in, rather than competing or conflicting (Levett, 1998). In the area of climate policy the perspectives of different scientific traditions may express different aspects of sustainability, with narratives representing different views on the role of emission trading versus national climate policy and on the role of stakeholders in shaping the policy debate (Miller, 2000).

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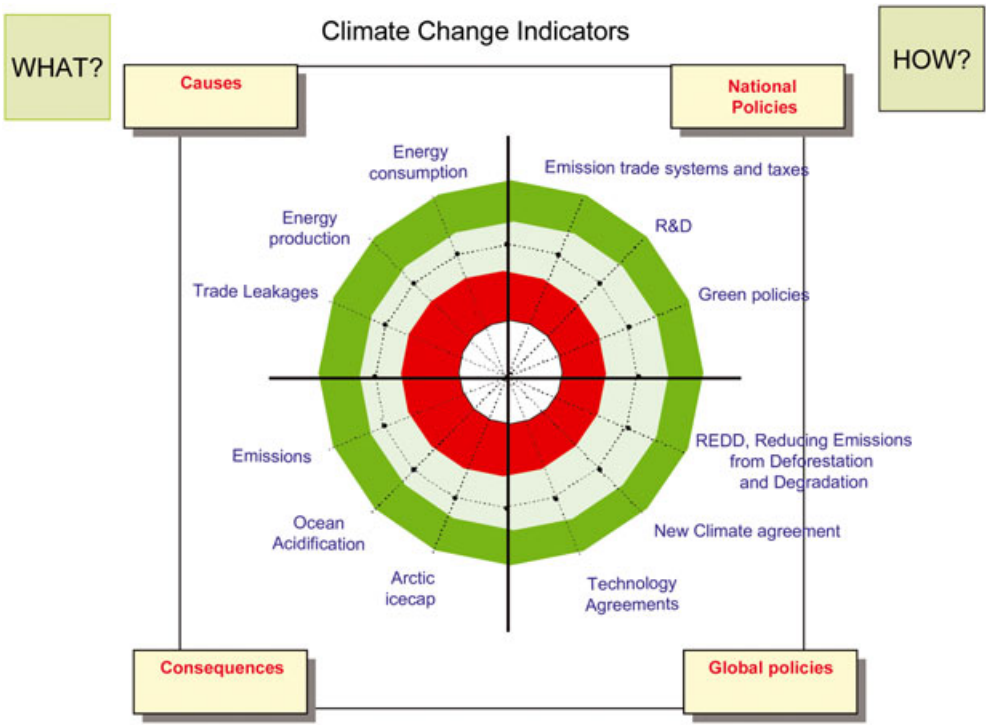


Figure 2. Different perceptions of the climate change problem and indicators for climate policy

Integrated Assessment, Single and Double Loop of Learning, and Narratives

An example of organizational learning for improving the sustainability assessment processes is given by the model of single loop and double loop learning (Argyris and Schon, 1978). Figure 3 illustrates the difference between single loop and double loop learning in the context of production and assessment of SDIs. The single loop emphasizes the process of learning by "following the rules". The double loop implies also changes in the rules themselves and makes explicit the arguments for these changes. The idea of the single-loop learning model is that feedback from the evaluation of the proposed SDI set is addressed towards the expert community only and involves technical issues of how the analysis is performed, whereas the double-loop learning model addresses the more fundamental underlying value assumptions and the purpose of developing the SDI set, and discusses its use for policy within deliberative processes. The double-loop feedback mechanisms point to the potential benefits from enhanced cooperation between the processes of policy-making and sustainability assessment and the technical development of indicators. The single-loop feedback mechanism, from D to C in Figure 3, may involve a call from policy-makers to the relevant ministry for clarification or supplementary information or analysis, without addressing the fundamental

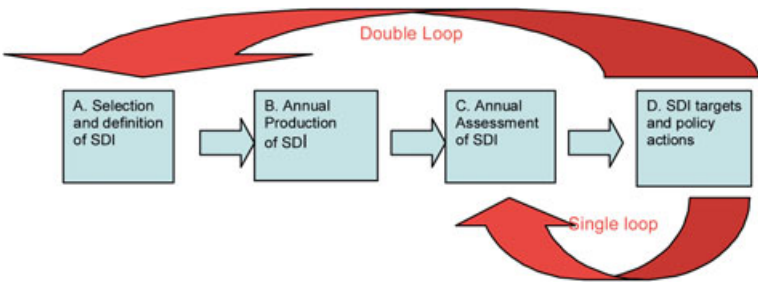


Figure 3. Double-loop and single-loop feedback process for production and assessment of SDIs

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basis for selection of indicators in terms of the purpose. The double-loop learning mechanism, from D to A, points to the situation where the feedback from the SDI targets and policy actions in the policy-making process (D) to the statistical development of indicators (A) results in possible redefinitions or refinements of the SDI set, or selection of new indicators, as well as the involvement of stakeholders.

The narrative of sustainability cannot be considered only from an economic or ecological point of view, as the issues of economic–ecological integration, and inter-generational and intra-generational equity are of fundamental importance, with large implications for understanding of sustainability and responsibility (Munda, 1997; Baumgärtner *et al.*, 2008). Sustainability indicators can contribute to express the relationship between purpose, learning and governance (Hezri and Dovers, 2006). An example is the recently developed Nature Index for Norway, which has been introduced as a sustainability indicator, with the potential of extending the communication between experts and policy-makers to a learning process involving stakeholder participation (Certain *et al.*, 2011; Aslaksen *et al.*, 2012).

In the context of climate policy, a double-loop learning process would typically address the questions: How is the need for a more ambitious climate policy perceived by different societal interests? How may different societal perspectives contribute to the framing of national climate policy and the sustainability discourse? It would be of great interest to survey and evaluate the experiences of countries where various elements of sustainability information and policy have been integrated through the assessment process, providing a base for double-loop learning.

It is widely recognized that sustainability indicators are contextually grounded and that such contextuality needs to be explicated, in particular in terms of how the sustainability problem is perceived, interpreted and framed (National Research Council, 1999; Hukkinen, 2003a; Hukkinen, 2006; Lawn, 2006). The concept of narrative can be useful for describing how different societal perspectives on sustainability express their priorities for selection of indicators and choice of policy (A and D in Figure 3). Using narratives in science–policy communication is a tool for describing and organizing information of high complexity (Allen and Giampietro, 2006; Giampietro *et al.*, 2007). Each narrative provides a particular context or framing for data, indicators and policies, serving to remind that the interpretation of sustainability by various groups in society is crucially dependent on their positions, values and interests. SDI sets are quantitative expressions of narratives.

Narratives are formulated "to explore the alternative choices that might lead to feared or hoped-for futures" (Cronon, 1992, p. 1368). Narratives have a long history in the philosophy of science: "Aristotle reminded us, so long ago, narrative is among our most powerful ways of encountering the world, judging our actions within it, and learning to care about its many meanings" (Cronon, 1992, p. 1375). Expressing implicit values is crucial in order to ensure that sustainable development is not only perceived in terms of the objectives of economic growth, national security and material consumption (Luke, 2005). The concept of narratives is used by the OECD Statistical Directorate in their webpage on "Statistical narratives" (OECD, 2010). A similar example is the "postulates" employed by Swiss Statistics (2010), which are qualitative statements of sustainability, where it is emphasized that an indicator should tell a story.

Giampietro and Sorman (2009) suggest a general framework for clarifying how the whole system of developing official statistics is conditioned by interaction processes between scientists, statistical experts, politicians, interest groups and the general public, as outlined in Figure 4. Official statistics serves a multitude of purposes, with users-representing different societal, professional and political interests and demands for statistical information. Sustainability indicators can be seen as the result of a process of societal and political interaction between established institutions and the rest of society, where national statistical offices have a key role. Although decisions on statistical definitions and procedures are made by statisticians, the compilation of statistics and sustainability indicators is not only a selection of facts in some technical sense, as the choices involved are conditioned by societal interests, and it is crucial to clarify implicit normative values embedded in the data-generating processes. The idea of the "multipurpose grammar" reflects that different groups of society have different demands for statistics, expressed not only in data but in their narratives of sustainability, and the expression of conflicting interests gives rise to competing narratives of sustainability.

The perspective expressed in Figure 4 is also considered by the Stiglitz report, taking into account the interrelationship between the role of national statistical offices and other institutions of society in providing consensus knowledge as a basis for policy (Stiglitz *et al.*, 2009). The Stiglitz report recognizes the complex normative questions involved in sustainability assessments: "Measuring sustainability also entails prior responses to normative questions. In this respect too, it strongly differs from standard statistical activity [. . .] Making choices in this respect goes once again far beyond the normal job or normal responsibility of the statistician; they can help clarify the

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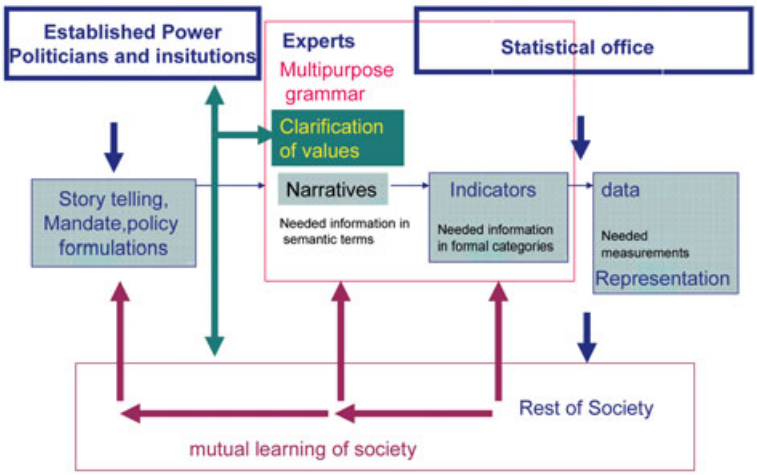


Figure 4. A general model for policy, experts and public interaction in the development of indicators [*Source*: Adapted from Giampietro and Sorman (2009).]

options or help implementing correctly the index once the choices have been made, but they can in no way fully assume the definition of objectives" (Stiglitz *et al.*, 2009, p. 264). The normative element in defining and measuring sustainability not only involves predictions of the future, but also different views of "what will really matter tomorrow for us or our descendants" (Stiglitz *et al.*, 2009, p. 242).

While the single loop approach typically will select indicators based on available models and data, the double loop approach will aim at explicating the context and the alternative narratives that reflect conflicting social interests that need to be taken into account to guide policy choices. In this approach, the selection of data will be grounded in a context where different societal interests are expressed in a process of "mutual learning of society" towards policy for sustainability.

Using the modelling-relation theory (Rosen, 2000) the process of selecting narratives for sustainability assessment may be combined with the development and application of sustainability indicators in an integrated five-step process (Figure 5). What we emphasize here is that the call for quantitative indicators may overshadow the importance of expressing the policy targets in words, i.e. in narratives reflecting values and interests. The novelty of this approach consists in placing the standard procedures of data gathering, sustainability assessment and policy application (steps 3–5) into a context of explicit choice of narrative and external reference (steps 1 and 2). The crucial question is: What is the perception of the problem we want to structure?

Cronon (1992) explains the contextuality of perception by different narratives of the "conquest of the West", perceived as development of virgin territories or as ecological disaster for the indigenous residents. The importance of perception can be illustrated by the example of an audience viewing a photo of an armed man, seen as a freedom fighter or a terrorist, depending on the individual context of perception, giving rise to different explanatory models and policy prescriptions. The contextuality of perception has large implications for the policy framing of sustainability. In terms of climate policy, as indicated in Figure 2, public support for climate-friendly technology may be perceived as expression of the environmental responsibility of rich countries or as a waste of public funds. Similarly, in terms of indicators for biodiversity policy, wolves can be perceived as iconic symbols of charismatic animals of the wilderness, representing conservation responsibility, or as a danger for children at play outdoors, lambs grazing on mountain pastures and the traditions of Sámi reindeer herding.

Uncertainty and Sustainability

Fundamental uncertainties arise from the unpredictability of natural systems, as well as social systems, and the interactions between them. The very nature of complex adaptive systems involves at least two non-reducible sources

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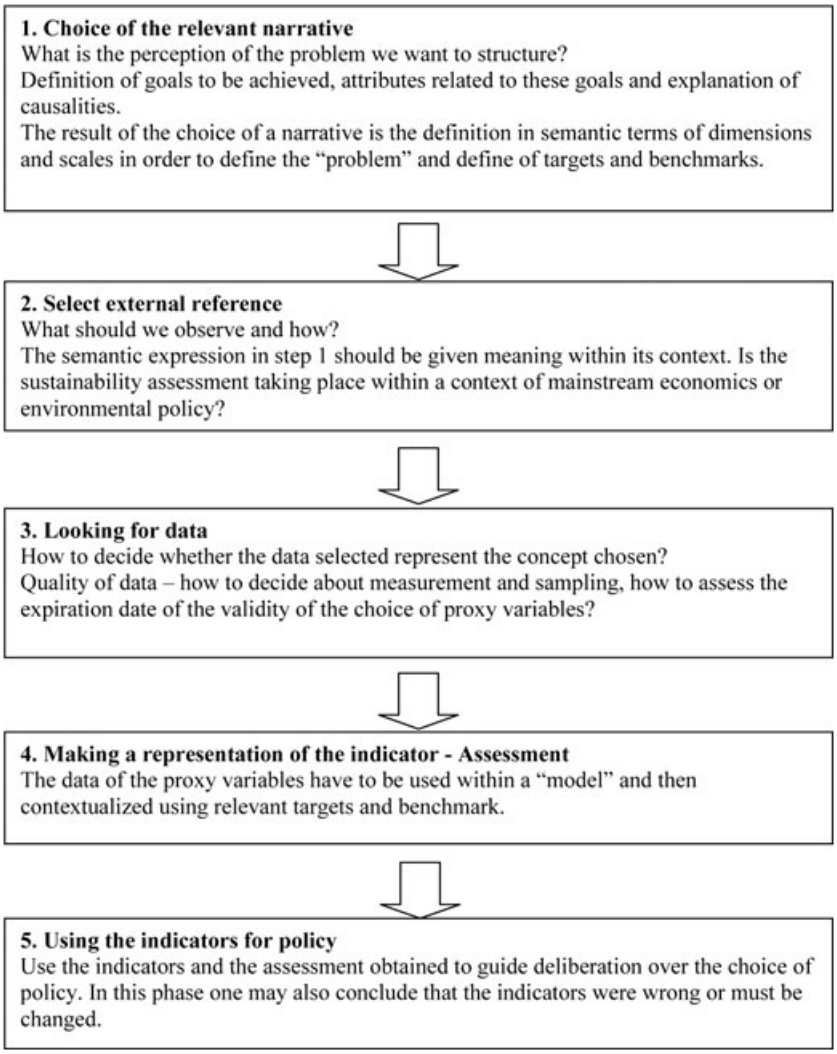


Figure 5. Selection of narratives for sustainability assessment as the five steps in modelling-relation theory

of uncertainty. (1) Their nature of "becoming systems" (Prigogine, 1978) entails that any quantitative representation based on a given set of measurable attributes sooner or later will become obsolete because of systems-evolutionary changes introducing new definitions of relevant attributes and issues (Georgescu-Roegen, 1976). (2) Their organization of multiple hierarchical levels requires the adoption of different scales for their perception and representation (Simon, 1962; Allen and Starr, 1982) and makes it impossible to adopt just a single mathematical model, no matter how complicated (Rosen, 2000; Giampietro *et al.*, 2006). Perception uncertainty regarding choice of narrative, anticipation uncertainty regarding choice of model and implementation uncertainty regarding choice of policy are factors challenging the application of statistical models for sustainability policy (Knight, 1964). In addition, we have to take into account the forward-looking uncertainties, the concept of "emergent uncertainties", a new type of evolution, a fundamental shift, and a more fundamental form of "ignorance" referring to the "unknown unknowns", because such categorizations are an invaluable help in structuring the narratives about uncertainty (Wynne, 1992; Bruun *et al.*, 2002; Renn, 2008). The 'uncertainty monster' refers to the confusion and ambiguity encountered when having to deal with the uncertainty between knowledge and ignorance, realizing that the scientific knowledge depends on value choices that blur the distinction between objectivity and subjectivity, facts and values, and prediction and speculation, and that the relationships between science and policy depend on the

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particular context (van der Sluijs, 2005; Curry and Webster, 2011). Forward-looking indicators are required, reflecting that the idea of precaution should be implicit in the notion of sustainable development.

The Stiglitz commission recognizes that uncertainty is normative, as they question how measures established today may be used to predict the valuations of future generations in situations that may have become very different:

> "It could be argued that our descendants may become very sensitive to the relative scarcity of some environmental goods to which we pay little attention today because they are still relatively abundant, and that this requires that we immediately place a higher value on these items just because we think that our descendants may wish to do so" (Stiglitz *et al.*, 2009, p. 75).

Forward-looking indicators are required, reflecting that precautionary approaches need to be at the core of policies for sustainable development.

Post-normal science has been conceptualized as a new perspective on the science–policy communication in response to the complexities created by the fundamental uncertainties of critical environmental problems and the inadequacy of scientific as well as political institutions in responding to these problems (Funtowicz and Ravetz, 1990). Post-normal science can be defined as the extension of scientific practice into situations when scientists take into account the intertwined relationships between facts and values, the possibility of catastrophic decision-stakes, the legitimate plurality of conflicting interests and ethical complexities, beyond what is usual in normal scientific practice. Moreover, the idea of post-normal science is that involvement and participation of stakeholders and citizens may contribute to improve the quality of the policy deliberations (Funtowicz and Strand, 2007, 2011). The double-loop learning process is a valuable response to a situation of complexity, as it encourages a continuous evaluation and adjustment of data sets, indicators, assessment results and policies:

> "Measuring sustainability differs from standard statistical practice in a fundamental way: [. . .] What is needed are projections, and not only projections of technological or environmental trends, but also projections of how they will interact with socio-economic or even political forces. Stated as such, the challenge is extreme. In practice, ambitions will remain much lower, i.e. just providing numbers identifying a risk of unsustainability under the continuation of current trends or behaviours. But even that task remains a considerable one that goes much beyond the normal job of statisticians and/or of economists. It requires a much broader set of expertise than is the case for usual accounting activities" (Stiglitz *et al.*, 2009, pp. 263–264). Path dependence has been identified as a challenge for SDI development, not just to foresee how resilient the sustainable path is, but also how resilient the unsustainable paths are (van den Bergh *et al.*, 2007; Hukkinen, 2003b).

A key property of any sustainability indicator is its ability to show changes over time. Indicator-based sustainability assessment needs to address its historic development, as well as its possible continuation into the future or relevant "early warnings" of emerging threats to sustainability, in order to evaluate the usefulness of the indicator for the assessment of whether society is on its way toward a sustainable development path. A classical type of sustainability analysis is to apply energy intensities to illustrate how energy production and consumption varies over time in different societies and in different environments. As an application of the idea of bioeconomics from Georgescu-Roegen (1971), Ulgiati *et al.* (2008) have suggested the approach of Multi-Scale Integrated Analysis of Societal and Ecological Metabolism (MSIASEM) to analyse internal complexities of the so-called metabolism of a social–environmental system on different levels and scales (Giampietro *et al.*, 2012).

Even the selection of measurement scale is not a trivial choice. Absolute changes in variables may provide easy interpretations in cases where the indicator may be directly linked to fundamental responses to processes in physical or environmental systems, or threshold values. Relative change (percentage change) may be the most relevant way of presenting changes in indicators over time.

It is important to be aware of possible confusion when dealing with different types of change when there is correlation between the variables. For example, an increase in the efficiency of using energy – measured as a decrease of the ratio of energy in MJ per € of GDP – may have the effect of increasing the value of per-capita GDP. For this reason it is important to adopt energy accounting protocols based on a combination of extensive (such as total population or total GDP) and intensive variables (such as per-capita GDP or per-capita primary energy

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consumption) to avoid adopting misleading models, for example that a decrease in the ratio MJ/€ implies a decrease in energy used in a society. In the case that the reduction in the ratio MJ/€ is compensated by a more than proportional increase in the value of per-capita GDP or by an increase in population, the effect can be opposite to that indicated by such a simple model (Giampietro *et al.*, 2012).

The assessment of trends in indicators is a central issue in sustainability assessments. One approach is to argue that the relevant scope of trend assessment would be to consider if a specific indicator describes a trend that is reasonably characteristic of a path in the direction of sustainable development or not. This step in the assessment process certainly requires a clarification of societal objectives in qualitative terms, although policy targets in quantitative terms may not have been specified.

A formal analysis of trends in indicators may be considered as part of a sustainability assessment, based on models or correlations, in terms of economic or econometric modelling or other types of modelling efforts. Decomposition analysis is primarily used as alternative to model analysis to analyse how a set of explanatory factors have contributed to changes in the indicators over time (Vehmas *et al.*, 2008). Formal and less formal techniques of foresight assessment may complement each other within the same context, in order to understand issues of complexity, uncertainty and surprises as they facilitate reflecting on the future and help anticipate where surprises might come from, even though it may not be possible to predict them (EEA, 2011).

Assessing the Overall Picture given by the SDI Set

For policy purposes, the issue is assessment of the overall picture given by the SDI set, in order to address the question: To what extent does the set of all indicator assessments give a signal about whether society is moving towards sustainable development or not? Some composite indicators have received much attention, most notably the Human Development Index (UN), Environmental Sustainability Index (World Economic Forum), Dashboard of Sustainability (EU) and Ecological Footprints among others (Moldan *et al.*, 2004). Saisana and Tarantola (2002) summarize some arguments for and against (Figure 6). Composite indicators may highlight interesting differences between countries in ways that contribute to improving knowledge on sustainable development. However, they can seldom be used as a basis for implementing specific policy measures.

Aggregation of incommensurable value dimensions may conceal differences in underlying indicators rather than clarifying them (Saisana *et al.*, 2005; Saltelli, 2007; OECD, 2008; Paruolo *et al.*, 2011). A composite index may neglect or disguise serious environmental problems and increase the difficulty of identifying proper remedial action: "In addition to raising technological issues, measuring sustainability with a single index number would confront us with severe normative questions. The point is that there can be as many indices of sustainability as there are normative definitions of what we want to sustain" (Stiglitz *et al.*, 2009, p. 75). For example, a measure of "green GDP", where gross domestic product (GDP) is extended with a measure of environmental quality, is less useful if growth in GDP overshadows environmental degradation.

Pros	Cons
• Can summarise complex or multi-dimensional issues in view of supporting decision-makers. • Easier to interpret than trying to find a trend in many separate indicators. • Facilitate the task of ranking countries on complex issues in a benchmarking exercise. • Can assess progress of countries over time on complex issues. • Reduce the size of a set of indicators or include more information within the existing size limit. • Place issues of country performance and progress at the centre of the policy arena. • Facilitate communication with general public (i.e. citizens, media, etc.) and promote accountability.	• May send misleading policy messages if they are poorly constructed or misinterpreted. • May invite simplistic policy conclusions. • May be misused, e.g., to support a desired policy, if the construction process is not transparent and lacks sound statistical or conceptual principles. • The selection of indicators and weights could be the target of political challenge. • May disguise serious failings in some dimensions and increase the difficulty of identifying proper remedial action. • May lead to inappropriate policies if dimensions of performance that are difficult to measure are ignored.

Figure 6. The pros and cons for composite indicators [*Source*: Saisana and Tarantola (2002).]

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Another problem is the issue of scale and time differences. The links between indicators must be considered, as increases in one indicator may necessarily lead to decreases in another. At the same time the indictors must clarify both the internal and the external constraints of the system (Giampietro *et al.*, 2012). Composite indicators may have their cons, but on the other hand, how can multi-dimensional sustainability development indicators be rescued from the 'sea' of the infinitely possible? To ensure that diversity is represented in the expression of priorities, specification of objectives and targets for sustainability, and assessment of whether the actual development is towards sustainability or not – as expression of the aspirations of society – it is crucial to extend the sustainability assessment from an expert-driven process to a participatory process representing a diversity of values. Again, the goal is not so much to create a certain final set of technical indicators, as it is to engage a range of stakeholders in society in discussions about what it is that we value, where we want to develop as a society, how we should handle uncertainties, and what that means to us and our descendants with respect to a commitment to the present, reflected in political action. An SDI set can help guide deliberation and assist policy-makers in accepting compromises over societal goals. A deliberation process seeking compromises may contribute to more transparent decision-making – "you know what you give away" – in contrast to trade-offs that are not transparent if value assumptions are not made explicit.

Whether the perceived urgency of environmental problems warrants increased public attention can always be questioned. To some extent the complexity of society calls for new areas of deliberation in addition to those of a representative democracy, involving an extended form of consultations and public participation. An SDI set clearly has a role to play as input in participatory processes where different types of knowledge relevant for different societal interests may enhance the knowledge basis for setting policy targets for sustainable development. Knowledge integration can be seen as an important element in institution building (Hukkinen, 2008). However, if SDI sets only become a platform for dialogue or debate, the result for sustainability might just as well be negative as positive. The specificity of sustainability calls for specificity of stakeholder participation to avoid tyrannies of vociferous minorities or non-committed majorities, and a blanket call for participation clearly may be unwarranted (Collins and Evans, 2002; Hukkinen, 2008).

Objectives and Targets: SDI Applied for Policy

If narratives representing different positions in sustainability policy are clearly expressed, it should be possible to identify objectives and targets. The 2009 monitoring report on the EU sustainable development strategy distinguishes between objectives and targets in the SDI pyramid (Eurostat, 2009): the top level of the pyramid comprises the overall objectives, representing widely used indicators with a high communicative and educational value. The second level comprises the operational objectives and targets, where the targets often will be expressed quantitatively. The third level comprises actions and explanatory variables. Underlying and supporting the SDI pyramid are the contextual indicators.

With regard to overall sustainability assessment, the question remains as to how large a change in an indicator is necessary to conclude whether development points towards sustainability or not. Short-term fluctuations must not be mistaken for changes in long-term trends. Indeed, a one-year dip in, for instance, climate emissions, as experienced in 2010 due to economic recession, does not indicate a new trend towards sustainable development.

The assessment procedure would be enhanced if the main indicator could be supported by indicators showing which policy instruments could be used to achieve the target and if they are used as planned. It may then be possible for politicians, scientific experts and citizens to evaluate the overall policy regarding national emissions. Levett (1998, p. 292) suggests that, "A handful of resonant headline indicators of sustainable development are fine - provided that we *also* collect and make available publicly enough relevant contextual and supporting material to allow anyone interested to judge whether the headline indicators actually mean what they appear (or purport, or are claimed by politicians) to mean. [. . .] We can think of resonant indicators as the mountain peaks visible above a layer of cloud, each buttressed and supported by a 'pyramid' of more technical indicators. This model can explain how what are sometimes seen as two completely different (and competing) kinds of indicator are in fact connected [. . .]."

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"Measuring sustainability raises an additional difficulty in an international context. The question is not exclusively to assess relative sustainabilities of each country taken separately. The problem is global, at least in its environmental dimension. In that case, what is at stake is rather the contribution of each country to global sustainability or unsustainability" (Stiglitz *et al.*, 2009, p. 264). This is obviously the case with climate policy. It is widely accepted that the developed countries must take more of the burden of emission reductions than developing countries, but the question is how much more. This is a question with answers in the domain of ethics and politics. As Jamieson (1992) puts it, the problems associated with climate change are not purely scientific but also concern how we ought to live and how humans should relate to each other and to the rest of nature. Hulme (2007) argues that in order to make progress about how we manage climate change it is necessary to have debates about wider social values. The problem is, however, that such discussions often masquerade as disputes about scientific truth and error. Possibly, one could more easily reach some agreement about the ethical principles for an international agreement than agreements on scientific truth or error.

An ethical approach could for instance be to hypothetically argue along the lines of a Kantian categorical imperative and suggest that all countries should act as if a comprehensive climate agreement had been achieved (Greaker *et al.*, 2012). In principle, if all countries did that, we would be closer to a sustainable path. The problem is of course that each country will articulate their responsibilities under such a hypothetical agreement in different ways, thus presenting different policies. If such hypothetical deliberations were to be taken seriously, it would, however, be a need for more comprehensive indicators of climate policy, expressing different ways of contributing to reduced emissions, such as investments in climate research projects, contribution to projects in developing countries or other types of contributions to international climate policy initiatives. This position reflects the ethical imperative that rich countries have an obligation to carry a larger burden. Indeed, indicators may be selected also to represent alternative narratives as to how the society may adapt to climate change and how well it manages to prepare for large and drastic changes (O'Brien, 2011). Nevertheless, bringing up ethical considerations could contribute to the important public deliberations on sustainability. As Jamieson (1992) argues: "One of the most important benefits of viewing global environmental problems as moral problems is that this brings them into the domain of dialogue, discussion, and participation. Rather than being management problems that governments or experts can solve for us, when seen as ethical problems, they become problems for all of us to address, both as political actors and as everyday moral agents."

It is generally observed that even if large resources are used on establishing SDI sets, their effectiveness in influencing actual policy and practices often remain limited (Parris and Kates, 2003; Bell and Morse, 2003; Pintér *et al.*, 2005). Policies to promote sustainable development require a new understanding of the path from statistics to policy. The challenge is to ensure that the indicators can become influential and useful in practice:

> "The literature deals mostly with the design of indicators. It focuses on developing a particular kind of output and pays little attention to the process of indicator production or the ways in which the numbers might become influential in practice" (Innes and Booher, 2000, p. 175).

Moreover, Innes and Booher (2000) emphasize the importance of ensuring that indicators are actually used in the policy process:

> "For indicators to be used there must be not just opportunity, but also a requirement to report and publicly discuss the indicators in conjunction with policy decisions that must be made. If this sort of required linkage is not made and followed, the indicators will never become part of the debate. If the indicators start moving in a direction that is politically problematic the producing agencies will stop publishing them" (Innes and Booher, 2000, p. 178).

Boulanger (2007) suggests a framework with three conceptual models of how SDI sets can be applied in the policy process:

- Rational–positivist model: objectives for sustainability are set by politicians, efficient policies are designed and the degree of success is measured by indicators in the assessment process.

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- Discursive–interpretive model: indicators are considered in terms of their usefulness for how the sustainability issues are framed, or expressed in a specific context, by narratives that represent different societal perspectives on sustainability.
- Strategic model: competition between different interest groups and stakeholders is emphasized, and the use of indicators in this context may be strategic in order to strengthen one's bargaining power (Boulanger, 2007).

The important issue is how useful the assessment is in a process of policy-making and that it addresses the question: "What is the purpose of the analysis?" The sustainability discussion needs to be taken out of the closed rooms of experts, scientists and politicians into an open debate, with sustainability assessment redefined from a technical process to a process of learning, participation and involvement.

An example of how model construction, use of the model and negotiation processes leading to international policy agreements are integrated is the RAINS model developed by IASSA in 1983–1984, inspired by the emerging Helsinki protocol on cutting sulphur emissions. In scientific terms the RAINS model was quite simple. The simplicity was part of the reason for its success, but the main reason was that the model served as a basis for political negotiations. In practice these negotiations influenced the input data, the political instruments selected and how the output was transformed to a national aggregate for each country. All this was part of a bargaining process towards the Oslo protocol of 1994. It was not the great scientific complexity of this model that made the success. It was the will to enter into political discussions. This again was based upon a strong support from the public experiencing the damages of acidification and demanding international action (Gough *et al.*, 1998).

Policy support for other environmental problems may draw on the learning experience of the application of the RAINS model. The main lesson from the RAINS model is related to the strong policy support for anti-acidification measures when the environmental consequences were experienced. This raises the issue of how to mobilize a process of collective action to deal with environmental problems. In the case of biodiversity loss, Hulme *et al.* (2011, p. 698) suggest a role beyond assessments for the new Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES):

> "[. . .] the goals of IPBES should be expanded. First, we should move beyond conventional scientific knowledge assessments that legitimize, almost exclusively, only peer reviewed material. Knowledge established across all scales (especially the knowledge of local and indigenous peoples) and validated in multiple ways must be eligible for inclusion in IPBES processes. [. . .] Second, we should link IPBES assessment results to decision-making at multiple spatial scales (including tackling biodiversity loss at the grassroots level). Both of these goals require all aspects of capacity-building, including empowerment of different kinds of actors, to be reflected in the structural design of IPBES."

Another example of sustainability discussions taken out of the closed rooms of experts is the Intergovernmental Panel on Climate Change (IPCC) process. In some ways the IPCC, with its extensive, decentralized network of scientists and politicians, has the potential to represent a considerable social innovation of science–policy communication (IAC, 2010). The IPCC has contributed to national and international discussions on climate research and policy among scientists, policy-makers, non-governmental organizations, businesses and the general public. As a response to criticism and increased public scrutiny, especially related to the Fourth Assessment Report, the IPCC has taken steps to harmonize the approaches and terminology used by scientists to describe uncertainty and to make its review processes more inclusive and transparent (IAC, 2010). It can be argued that the IPCC thus has promoted what could resemble a double-loop learning process, encompassing a wide range of perspectives and policy framings from different societal interests, although it has not sufficiently drawn out policy implications of their analysis, as their reports should "be neutral with respect to policy" (IPCC, 2011).

Sustainability assessment also requires a critical discussion of how the scientific knowledge-generating processes are organized. More and more knowledge is generated to promote technical efficiency and innovations. We have to some extent stopped thinking that people may change their lifestyles. Instead we find a technical fix as the answer to sustainability. Knowledge generation itself is controlled and communicated within a context of strong economic, political and professional interests. Uncertainties and non-acceptable outcomes are not necessarily brought to public attention. The main problem here is how to bring public control into the science–policy process. This goes for both

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public access to research results but also in the deliberation on how new knowledge is to be used in political decision-making. If these concerns were to impact on some of the science–policy questions, a new type of policy deliberation has to be organized. The existing forms for setting policy priorities have to be extended beyond government bureaucracy and standard political processes.

Concluding Remarks

Policies to promote sustainable development require a new understanding of the path from statistics to policy, where sustainability assessment is redefined from a technical process to a deliberative process of learning, participation and involvement. Deliberations in terms of expressing the aspirations of society may be a response to rescuing sustainability from the 'sea' of the infinitely possible. The challenges for the national statistical offices are different in the presentation of sustainable development indicators than in other type of statistics. The choice of narratives and how to describe these narratives in semantic terms is crucial. The perception of the context is vital in order to decide on the data representation. The selection of indicators is not a technical process, but includes normative value judgments. Assessment of sustainability needs to take place within learning institutions, with extended participation of societal interests, and with feedback between the development of statistical indicators and their use in sustainability policy.

The main findings of our discussion are first that improvements in sustainability assessment procedures are embedded in a social context where knowledge for sustainability policy is developed in mutual learning processes involving national statistical offices and other institutions and stakeholders. Secondly, the stepwise procedure of developing the narratives of sustainability (Figure 5) emphasizes that it is crucial to consider how the sustainability problem is perceived for the selection of indicators and their usefulness for policy. Finally, it is emphasized that the rationale for developing sustainability indicators and assessments is their usefulness for policy. The main message is that it is not only the overall quality of the indicators that matters, but rather how these data and assessments are deliberated in a political process reaching agreements for political action. Successful use of an SDI set means that a sustainable policy, expressed as targets and policy instruments, is negotiated in a political process focused both on long-term goals and short-term actions.

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